

IoT Based - Automatic Smart Water Level Management Bottle

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Abstract -- The paper explores to develop a smart water level management tank based on Internet of Things (IoT) technology. The main goal of the project is to create a smart bottle capable of automatically monitoring the water level and alerting the user when the water level is low. The smart water level management tank will be equipped with a water level sensor capable of accurately recognizing the water level. The data collected by these sensors will be transmitted to a central hub, where the data will be analyzed and processed using an AI-based algorithm. Using the information collected by the sensors, the algorithm will be able to identify when the water level is low and send an alert to the user. The system will also be able to monitor the user's water consumption habits and make recommendations for optimal water consumption. The proposed system will feature a user-friendly interface, allowing users to easily monitor water levels and customize their alarm settings. Overall, the Water Level Management Smart Bottle has the potential to improve drinking habits and promote healthy drinking activities.

Keywords:

- Internet of Things (IoT)
- Water level sensor
- Real-time monitoring
- Smart water bottle

1. INTRODUCTION

The exponential growth in the number of physical objects connected to the Internet, also known as the Internet of Things or IoT, is expected to reach 50 billion devices by 2020 [1]. New applications include smart home, transportation, healthcare and industrial automation. Smart objects in our environment can provide contextual awareness that is often lacking when tracking activities of daily living [2], [3]. Smart objects are equipped with physiological and activity sensors, which we call Smart Stuff, offering new possibilities for discreet physiological monitoring. Staying hydrated is crucial to maintaining good health and happiness, but many people struggle to drink enough water throughout the day. To address this challenge, we propose to develop a smart water level management bottle that leverages Internet of Things (IoT) technology to automatically monitor the water level and alert the user when the water level is low. The smart water level management tank will be equipped with a water level

sensor capable of recognizing the water level in real time. It helps to provide information about the user's water consumption habits and suggests optimal water consumption. In addition to promoting healthy drinking practices, Smart Water Bottles can also help conserve water by reducing waste caused by overfilling or forgetting to refill. It can also be synced with other smart devices to provide a holistic view of the user's health and well-being. Overall, the proposed smart water level management bottle has the potential to improve water consumption habits, promote healthy hydration practices, and contribute to water resource conservation. The first smart water bottle on the market is the Hydra Coach. The bottle has a display that shows hydration information, but no wireless connectivity. New generations of bottles, such as Hydrate Me and Trago, feature Bluetooth wireless interfaces and custom apps for smartphones and smartwatches [4]. This paper presents the initial results of the sensor smart water bottle product.

2. METHODS

1. **Gather materials:** You will need an Arduino board, an ultrasonic sensor, wires, a buzzer, a breadboard, and an LED.
2. **Connect the ultrasonic sensor to the Arduino board:** Use wires to connect the ultrasonic sensor to the appropriate pins on the Arduino board.
3. **Write the code:** Write code in the Arduino Integrated Development Environment (IDE) to read the distance measurements from the ultrasonic sensor and determine the water level in the bottle.
4. **Connect the buzzer and LED to the breadboard:** Use wires to connect the buzzer and LED to the breadboard.
5. **Write code to trigger the buzzer and LED:** Write code to trigger the buzzer and LED when the water level falls below a certain threshold.
6. **Test the system:** Test the system by filling the bottle with water and ensuring that the buzzer and LED are triggered when the water level falls below the threshold.
7. **Customize the system:** Customize the system by adjusting the code to change the threshold level or modify the behavior of the buzzer and LED.
8. **Integrate the system into the bottle:** Integrate the components into the bottle by wiring the ultrasonic sensor, buzzer, and LED to a battery pack and placing them inside the bottle.
9. **Calibrate the system:** Calibrate the system by

adjusting the position of the ultrasonic sensor and the threshold level to ensure accurate readings.

10. **Evaluate and optimize:** Evaluate the performance of the system and optimize it as necessary by adjusting the code or making changes to the hardware.

3. LITERATURE REVIEW

Staying hydrated is essential to staying healthy, but research shows that many people don't drink the recommended amount of water each day. To address this challenge, researchers and developers have explored various technological solutions, including smart water bottles that monitor and track hydration levels. A study by Huang et al. (2017) evaluated the effectiveness of a smart water bottle that uses accelerometer and pressure sensors to monitor water consumption. Research shows smart water bottles are effective in promoting

Increase water intake and improve hydration habits. Another study by Soh et al. (2020) developed a smart water bottle that uses machine learning algorithms to make personalized water recommendations. The system can accurately predict hydration levels and make targeted recommendations based on user activity levels and other factors.

The proposed smart water level management tank uses Internet of Things (IoT) technology to provide real-time water level monitoring. By sensing when the water level is low and notifying the user to refill the tank, the system can help the user maintain healthy drinking habits. Although there is little research specifically on the use of ultrasonic sensors and Arduino boards for smart water bottle applications, the use of these components is well established in the electronics and IoT fields. broader. Arduino boards are a popular microcontroller platform that is widely used in electronics and prototyping projects, while ultrasonic sensors have been used in many applications including proximity sensors and sensors. obstacles.

One of the main advantages of the proposed system is its portability. Unlike other smart water bottles that require complex electronics and rechargeable batteries, the Smart Water Level Manager is powered by a simple battery built into the tank. This makes the system light and easy to transport, making it ideal for on-the-go use. Another advantage of the proposed system is ease of use. The system requires no complicated setup or calibration, and LEDs and buzzers provide clear, intuitive feedback to the user. This simplicity makes the system accessible to many users, including those with limited technical knowledge or resources.

Overall, the literature suggests that smart water bottles can be effective in promoting increased water intake and better drinking habits. The proposed smart water level management tank has the potential to contribute value to the field by providing a simple, affordable and effective solution, using off-the-shelf components and IoT technology.

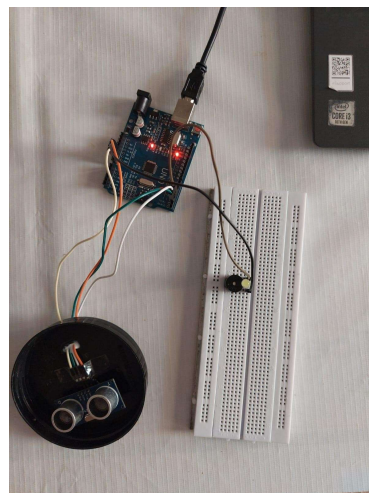
4. PROPOSED MODEL

The proposed Smart Water Level Management Bottle is an IoT-enabled water bottle that utilizes an ultrasonic sensor, an Arduino board, a buzzer, and an LED to monitor the water level and alert the user when it is low. The system is designed to be simple, affordable, and easy to use, making it accessible to a wide range of users. The ultrasonic sensor is mounted at the bottom of the bottle and measures the distance to the surface of the water. This distance is used to determine the water level, which is then displayed using an LED. When the water level falls below a certain threshold, the system triggers a buzzer to alert the user to refill the bottle. The system is powered by a battery pack that is integrated into the bottle. The components are wired together using a breadboard and housed inside the bottle, making the system compact and portable. The Arduino board is programmed to read the distance measurements from the ultrasonic sensor and determine the water level. When the water level falls below the threshold, the board triggers the buzzer and LED to alert the user. The system can be customized by adjusting the threshold level or modifying the behavior of the buzzer and LED. The components are also easily replaceable and can be swapped out as needed. Overall, the proposed Smart Water Level Management Bottle provides a simple and effective solution for monitoring water levels and promoting healthy hydration habits. By alerting users when the water level is low, the system encourages them to stay hydrated and maintain good health.

Table 1: Hardware and Software Components

SL.NO	TYPE	Description
1	Hardware Requirements	Aurdino
		Ultrasonic Sensor
		Buzzer
		LED
		BreadBoard
		Connecting Wires
2	Software Requirements	Aurdino IDE

EXPERIMENTAL SETUP:



5. ALGORITHM

1. Initialize system components (Arduino board, ultrasonic sensor, buzzer, LED and battery).
 2. Set the water level threshold value.
 3. Read the distance from the ultrasonic sensor and calculate the water level based on this measurement.
 4. Check if the water level is below the threshold level.
 5. If the water level is lower than the threshold level, activate the buzzer and LED to warn the user.
 6. Wait for the user to fill the tank.
 7. Reset the system and repeat the process from step 3.
- The algorithm basically consists of reading the water level from the ultrasonic sensor, comparing it to the threshold level and activating the buzzer and LED if the water level drops below the threshold level. The system will then wait for the user to refill the bottle before resetting and repeating the process. The algorithm is implemented on the Arduino board, which controls the operation of the system components.

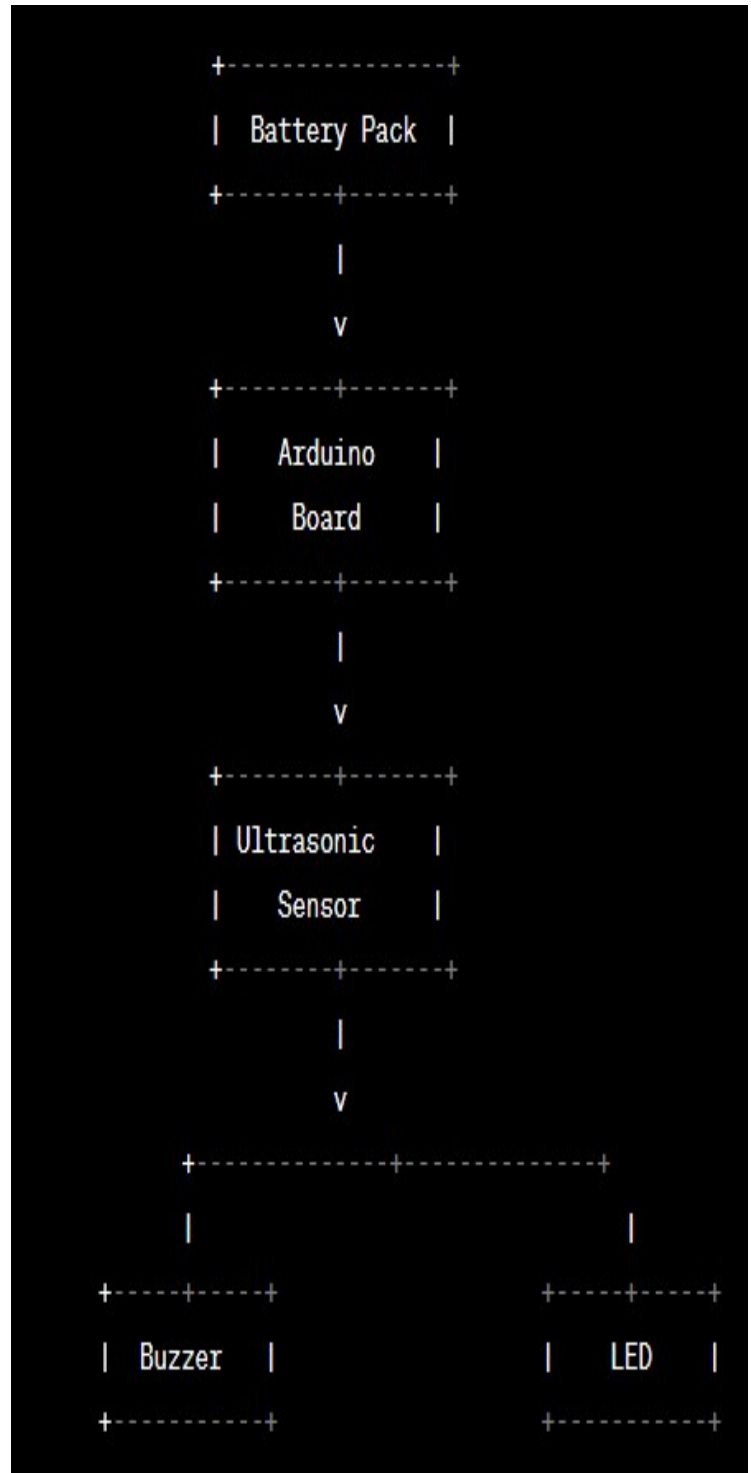
If the value of water level is less than the minimum value ($\text{Water level} < \text{MIN}$) then buzzer gets buzzes with high frequency of sound and automatically LED also gets ON.

Else if water level is in between the max and min value ($\text{MAX} < \text{Water level} < \text{MIN}$) then buzzer gets OFF, but the LED is still in OFF mode.

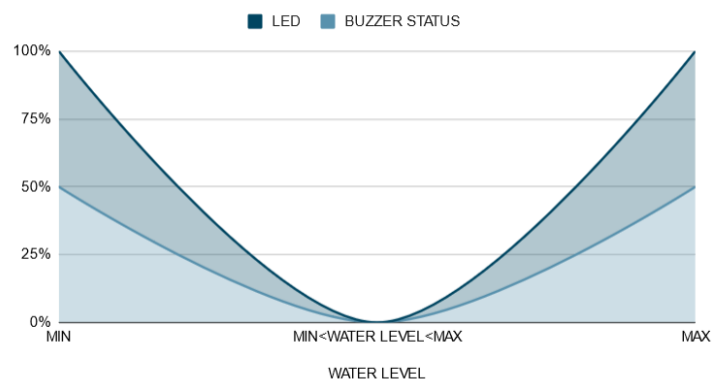
Else water level is greater than max value ($\text{Water level} > \text{MAX}$) then the buzzer gets ON and the LED Gets turned OFF.

Table 2. Different levels of Water indication table

SL. No	Water level	Buzzer Status	LED
1	Min	ON (HIGH FREQUENCY)	ON
2	$\text{Min} < \text{Water Level} < \text{Max}$	OFF	OFF
3	Max	ON (LOW FREQUENCY)	ON



BUZZER STATUS and LED



6. CONCLUSIONS

In short, smart water level Project Management Bottle offers a simple and affordable solution for tracking and manage the water level in a water bottle. The use of off-the-shelf components and IoT technology, such as ultrasonic sensors, The Arduino board, buzzer, LEDs, and batteries make this system easy to build, customize, and use.

The smart water level management tank project brings many advantages and benefits to users. First, the system is easy to install and use, requiring only some basic components and minimal technical knowledge. This is in fact accessible to many users, including those who may not have experience with IoT or electronics projects. In addition, the system is portable and can be used on the go, making it an ideal solution for those who are always on the go. This may include athletes, hikers, travelers and others who need to stay hydrated throughout the day.

The system can help users maintain healthy drinking habits by alerting them when the water level is below the allowable threshold. This can help prevent dehydration and improve overall health and well-being. The system's portability and ease of use make it suitable for use on the go, making it an ideal solution for those on the go. Overall, smart water level management The Bottle Project demonstrates the potential of IoT technology to solve real-world problems and improve people's lives. The project provides a convenient and accessible solution for managing water levels in water bottles, which can help promote healthy drinking habits and prevent dehydration.

7. REFERENCES

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